Multiplying Two-Digit Numbers

Break both numbers into tens and ones. Four pieces, then add.

Tens times tens

$20 \times 30 = 600$

2 tens × 3 tens: 2 × 3 = 6,
and tens times tens are hundreds.

23 × 14: four pieces

	20	3
10	200	30
4	80	12

10 × 20 = 200
10 × 3 = 30
4 × 20 = 80
4 × 3 = 12

200 + 30 + 80 + 12 = 322

No picture? List the four parts

Tens × tens, tens × ones, ones × tens, ones × ones. Then add.

$31 \times 26 = 600 + 20 + 180 + 6 = 806$

Your turn!

	10	2
10	100	20
3	30	6

12 × 13 = 156

For teachers

Goal: two-digit × two-digit with the area model, then partial products (4.NBT.5). It is packet 56's model with both numbers split. Say “Tens times tens first: those are hundreds.”

Multiplying Two-Digit Numbers: Practice

Split both numbers. Four pieces. Add them.

Part 1 Tens times tens 2 tens × 3 tens = 6 hundreds.

1) 30 × 40 = 1,200	2) 20 × 70 = 1,400	3) 50 × 60 = 3,000
4) 80 × 30 = 2,400	5) 40 × 90 = 3,600	6) 60 × 20 = 1,200
7) 70 × 70 = 4,900	8) 20 × 50 = 1,000	9) 90 × 30 = 2,700
10) 40 × 40 = 1,600	11) 60 × 80 = 4,800	12) 30 × 50 = 1,500

Multiplying Two-Digit Numbers: Practice (continued)

Part 2 The area model The tens and ones are on the outside. Fill in each piece.

13) **24 × 13 = 312**

	20	4
10	200	40
3	60	12

200 + 40 + 60 + 12 = 312

14) **32 × 21 = 672**

	30	2
20	600	40
1	30	2

600 + 40 + 30 + 2 = 672

15) **15 × 42 = 630**

	10	5
40	400	200
2	20	10

400 + 200 + 20 + 10 = 630

Multiplying Two-Digit Numbers: Practice (continued)

Part 2 The area model (continued)

16) **43 × 26 = 1,118**

	40	3
20	800	60
6	240	18

800 + 60 + 240 + 18 = 1,118

Part 3 Split them yourself Write the tens and ones on the outside first.

17) **36 × 12 = 432**

	30	6
10	300	60
2	60	12

300 + 60 + 60 + 12 = 432

18) **27 × 31 = 837**

	20	7
30	600	210
1	20	7

600 + 210 + 20 + 7 = 837

Multiplying Two-Digit Numbers: Practice (continued)

Part 3 Split them yourself (continued)

19) **52 × 18 = 936**

	50	2
10	500	20
8	400	16

500 + 20 + 400 + 16 = 936

20) **41 × 35 = 1,435**

	40	1
30	1,200	30
5	200	5

1,200 + 30 + 200 + 5 = 1,435

Part 4 Write the four parts No picture this time.

21) **34 × 21**

30 × 20 = 600
4 × 20 = 80
30 × 1 = 30
4 × 1 = 4
600 + 80 + 30 + 4 = 714

22) **16 × 45**

10 × 40 = 400
6 × 40 = 240
10 × 5 = 50
6 × 5 = 30
400 + 240 + 50 + 30 = 720

Multiplying Two-Digit Numbers: Practice (continued)

Part 4 Write the four parts (continued)

23) **53 × 32**

50 × 30 = 1,500
3 × 30 = 90
50 × 2 = 100
3 × 2 = 6

1,500 + 90 + 100 + 6 = 1,696

24) **28 × 17**

20 × 10 = 200
8 × 10 = 80
20 × 7 = 140
8 × 7 = 56

200 + 80 + 140 + 56 = 476

Part 5 Times a tens number No help this time.

25) 34 × 20 = 680
26) 17 × 30 = 510
27) 46 × 40 = 1,840
28) 23 × 60 = 1,380
29) 58 × 20 = 1,160
30) 19 × 50 = 950
31) 62 × 30 = 1,860
32) 35 × 40 = 1,400
33) 71 × 20 = 1,420

Multiplying Two-Digit Numbers: Practice (continued)

Part 6 Two digits times two digits No help this time.

34) $23 \times 15 = 345$	35) $41 \times 12 = 492$	36) $36 \times 24 = 864$
37) $52 \times 13 = 676$	38) $18 \times 27 = 486$	39) $64 \times 11 = 704$
40) $33 \times 32 = 1{,}056$	41) $47 \times 21 = 987$	42) $26 \times 35 = 910$
43) $72 \times 14 = 1{,}008$	44) $15 \times 19 = 285$	45) $38 \times 43 = 1{,}634$

Multiplying Two-Digit Numbers: Practice (continued)

Part 6 Two digits times two digits (continued)

46) $54 \times 26 =$ **1,404**

47) $29 \times 31 =$ **899**

48) $61 \times 47 =$ **2,867**

Part 7 Estimate, then multiply

No help this time. Round each number to the nearest ten first.

49) 38×21 is about **800**

50) 49×32 is about **1,500**

51) 61×19 is about **1,200**

52) 27×42 is about **1,200**

53) 52×48 is about **2,500**

54) 73×29 is about **2,100**

Part 8 Multiply

55) $81 \times 16 =$ **1,296**

56) $82 \times 36 =$ **2,952**

57) $81 \times 44 =$ **3,564**

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

58) $32 \times 24 = 768$

59) $24 \times 15 = 360$

60) $58 \times 34 = 1{,}972$

61) $31 \times 33 = 1{,}023$

62) $38 \times 48 = 1{,}824$

63) $76 \times 25 = 1{,}900$

64) $35 \times 19 = 665$

65) $69 \times 26 = 1{,}794$

66) $27 \times 35 = 945$

67) $42 \times 22 = 924$

68) $45 \times 12 = 540$

69) $38 \times 29 = 1{,}102$

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

70) 81 × 34 = 2,754

71) 63 × 24 = 1,512

72) 25 × 35 = 875

73) 58 × 41 = 2,378

74) 28 × 17 = 476

75) 62 × 35 = 2,170

76) 88 × 23 = 2,024

77) 62 × 27 = 1,674

78) 56 × 13 = 728

79) 76 × 38 = 2,888

80) 73 × 49 = 3,577

81) 58 × 35 = 2,030

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

82) $18 \times 18 =$ **324**

83) $38 \times 16 =$ **608**

84) $29 \times 15 =$ **435**

85) $55 \times 22 =$ **1,210**

86) $36 \times 12 =$ **432**

87) $51 \times 38 =$ **1,938**

88) $75 \times 48 =$ **3,600**

89) $47 \times 31 =$ **1,457**

90) $76 \times 18 =$ **1,368**

Part 9 Remember? Problems from earlier packets.

91) $136 \times 4 =$ **544**

92) $47 \div 6 =$ **7** R **5**

Multiplying Two-Digit Numbers: Practice (continued)

Part 9 Remember? (continued)

93) **6,284 to the nearest thousand =** 6,000

94) $\frac{3}{4}$ **of 24 =** 18

95) **3,508 + 2,694 =** 6,202

96) $\frac{7}{10}$ > $\frac{3}{10}$

Part 10 Word problems Draw the area model if it helps.

97) A theater has 24 rows with 18 seats in each row. How many seats are there?

Number sentence **24 × 18 = 432**

Answer **432** seats

Draw or show your work

98) A baker makes 36 trays of muffins. Each tray holds 12 muffins. How many muffins is that?

Number sentence **36 × 12 = 432**

Answer **432** muffins

Draw or show your work

99) A book has 15 chapters of 23 pages each. How many pages does it have?

Number sentence **15 × 23 = 345**

Answer **345** pages

Draw or show your work

Multiplying Two-Digit Numbers: Practice (continued)

Part 10 Word problems (continued)

100) Each of 27 kids reads 30 minutes. How many minutes do they read in all?

Number sentence **27 × 30 = 810**

Answer **810** minutes

Draw or show your work

101) Challenge: 12 boxes of 24 pens and 15 boxes of 20 pencils. How many more pencils than pens?

Number sentence **15 × 20 = 300, 12 × 24 = 288, 300 − 288 = 12**

Answer **12** more pencils

Draw or show your work

Check-In: Multiplying Two-Digit Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 The area model

1) **19 × 22 = 418**

	10	9
20	200	180
2	20	18

200 + 180 + 20 + 18 = 418

Part 2 Tens times tens

2) **80 × 60 = 4,800**

3) **50 × 20 = 1,000**

4) **30 × 70 = 2,100**

Part 3 Multiply

5) **47 × 26 = 1,222**

6) **54 × 36 = 1,944**

7) **19 × 31 = 589**

8) **61 × 16 = 976**

9) **68 × 46 = 3,128**

10) **33 × 36 = 1,188**